

Evaluation Report for **TEBT-2023-0006**

Submission Title	Protocol: Testing the performance of INVITES-IN, a tool for assessing the internal validity of in vitro studies
Submission Type	Primary Study Protocol
Preprint DOI	10.5281/zenodo.10119424
Evaluators	Olwenn Martin, ORCID: 0000-0003-2724-7882 , handling editor, UCL Alexis Descatha, ORCID: 0000-0001-6028-3186 , reviewer, INSERM Rashmi Joglekar, ORCID: 0000-0002-9912-6628 , reviewer, UCSF 1 anonymous reviewer
Status	Accepted
Evaluation Date	23 November 2023
Evaluation Round	Editorial review round 2
Recommendation	Accept

Acceptance Note

This manuscript has been accepted for publication by the handling editor after assessment in 1 round of editorial triage and 1 round of peer-review.

During triage, the handling editor identified three minor issues with the initial submission, including with the informativeness of declaration of interests for some co-authors, and issues around the identification of study participants and inclusion of analyses (qualitative or quantitative) of the influence of participant characteristics on the outcomes of interest and the lack of a reference for ethical approval. These minor issues could be addressed following peer review, and the manuscript was advanced to peer-review.

During peer-review, the evaluators recommended that the methods be clarified, particularly in relation to the systematised review of the literature, statistical methods (include mathematical equations and code for power simulations) and the reporting of assessment of accuracy, that the authors consider to what extent their current understanding of the potential user community had considered those developing/validating in silico methods, and that the statistical analysis of

accuracy give due consideration to existing methods and common statistics applied when investigating the value of diagnostic tools.

The authors addressed these issues by adding the missing information for the data protection impact assessment, explicitly referring readers to the protocol describing the development of the tool and adding details in the text where appropriate (e.g. regarding relevant risk-of-bias domains), clarifying the expertise of end users and possible affiliations of participants, clarifying methods including adding equations and codes for statistical analyses and including a section highlighting the limitations of the current protocol. Some co-authors reviewed and amended their declarations of interest.

The following points of scientific interest were surfaced by the review process; powering statistical analyses of the impact of characteristics of participants would require a number of test-users and therefore resources that are unlikely to be made available for such performance testing.

A complete record of the evaluation reports for this submission can be found at <https://zenodo.org/doi/10.5281/zenodo.8338899>.

Peer-Review Evaluation, Round 1

Submission Title	Protocol: Testing the performance of INVITES-IN, a tool for assessing the internal validity of in vitro studies
Submission Type	Primary Study Protocol
Preprint DOI	10.5281/zenodo.8315605
Evaluators	Olwenn Martin, ORCID: 0000-0003-2724-7882 , handling editor, UCL Alexis Descatha, ORCID: 0000-0001-6028-3186 , reviewer, INSERM Rashmi Joglekar, ORCID: 0000-0002-9912-6628 , reviewer, UCSF 1 anonymous reviewer
Status	In Peer-Review
Evaluation Date	17 October 2023
Evaluation Round	Peer review round 1
Recommendation	Revise
Explanation	Most of the suggested changes are editorial in nature. These include some clarification of the methods, particularly in relation to the systematised review of

Submission Title	Protocol: Testing the performance of INVITES-IN, a tool for assessing the internal validity of in vitro studies
Submission Type	Primary Study Protocol
	the literature, statistical methods (include mathematical equations and code for power simulations) and the reporting of assessment of accuracy. Moreover, there may be some merit in checking through the language of a revised manuscript for clarity and considering including the more detailed description of the statistical methods in supplemental materials rather than the main body of the manuscript to improve the flow. Some suggestions may require more conceptual changes to specific parts of the methods proposed in this protocol. First, the authors should consider to what extent their current understanding of the potential user community has considered those developing/validating in silico methods. Finally, the statistical analysis of accuracy should give due consideration to existing methods and common statistics applied when investigating the value of diagnostic tools.

Summary of reviewer comments

This manuscript describes a protocol for user-testing of the performance of a risk-of-bias tool (INVITES-IN) for in vitro (cell cultures) studies. The methods include development of a gold standard and user testing comprising of practical exercises and a user survey. Specifically, the authors are interested in assessing the consistency of the tool (whether it yields the same results for different users), the expertise and effort needed to use the tool (assessor workload), the potential for errors (accuracy), and how easy it is to use (user experience).

Seriousness of issues identified with planned methods	The reviewers identified some important issues with very specific part of the protocol that if addressed could notably improve the usefulness of this user-testing exercise.
Potential additional work to overcome issues	Most of the additional work required to address the reviewers' comments is editorial in nature. The consideration of who should be recruited as the potential user group and adequate statistical analyses to test the accuracy of the tool may require some reconceptualization of these specific parts of the proposed methods.
Acceptability of characterising limitations instead of planning additional work	In line with the fact that this manuscript described a protocol, the tendency would lean towards addressing comments rather than describing the limitations of currently proposed methods. If this is deemed unrealistic, justification and discussion of the limitations should be given.

Structured assessment

1. Are the research questions important?

☐ Strongly disagree	☐ Somewhat disagree	☒ Neither agree nor disagree	☐ Somewhat agree	☐ Strongly agree
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Comments: The manuscript describes the protocol for user-testing of a risk-of-bias tool, the methods for development of which have been described in a previously published manuscript. The importance of the development of the tool was addressed in the latter. Reviewers did not comment on the importance of the research questions, and this may simply have been taken as a given.

Suggestions for addressing the comments: None

2. Are the proposed hypotheses logical, rational, and/or plausible?

☐ Strongly disagree	☐ Somewhat disagree	☐ Neither agree nor disagree	☒ Somewhat agree	☐ Strongly agree
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Comments: Overall, the objectives are clear, and the study is well-designed to address them. It may be useful for readers to be given more details about the protocol for the development of the tool without having to refer to the previously published protocol. An important question is related to assumptions about the composition of the user community. Specifically, the user community is as likely to consist of those developing or seeking to validate *in silico* methods, models or testing strategies. It is not clear whether these users have been considered in the recruitment of users testing the tool.

Reviewer 2

1. As reviewer knowing nothing about INVITES-IN (as not released to the public yet) but commenting on a protocol for its performance testing is *per se* a bit odd. But the study design appears to be well-structured and very detailed, with clear objectives and how they will be evaluated. The statistical part is in parts a bit unclear, but I guess of secondary importance as the exact number of minimal sample sizes should be considered only as guidance.
2. The manuscript states “*expected end users of INVITES-IN are persons preparing literature reviews including in vitro studies ...*”, but the main users of *in vitro* data outcomes (especially related to simple KIEs) are usually those involved in *in silico* work (QSAR, AOP, IATA, Read-Across etc.) in preparation for chemical hazard and risk assessments (and probably only those non-experimenters who really have to read the details). I am wondering how well these experts were included, as “*expected to have experience with in vitro methods*” is rather vague and could be misunderstood as “being involved in the lab and/or test system development”.

Reviewer 3

1. The manuscript does not clearly describe the signalling questions or the risk of bias (RoB) domains underlying this method, or how the signalling questions are rated to complete the RoB analysis. These steps should be clearly stated/defined either in the methods, or in the supplemental materials. Without this information, the reader cannot discern the appropriateness of the tool or the robustness of the overall method. For example, it would be concerning if a quantitative scoring method was used to evaluate the signalling questions and assess RoB. In a recent consensus report entitled "[The Use of Systematic Review in EPA's Toxic Substances Control Act Risk Evaluations \(2021\)](#)" domain (see Table 3) the National Academy of Sciences (NAS) stated that: "The reliance on numeric quality scores is problematic because scores do not distinguish between high- and low-quality studies, and the relationship between quality scores and an association or effect is inconsistent and unpredictable"...and "More generally, the use of numerical scoring in critical appraisal does not follow standards for the conduct of systematic reviews."
2. The RoB domains should include a domain for financial conflict of interest. This recent publication suggests that the INVITES-IN tool does not consider financial conflicts of interest as a RoB domain (see Table 3). While the RoB domains are not clearly listed in the current manuscript, it is critical that the authors consider financial conflict of interest as a key RoB domain, as recommended by the National Academy of Sciences, which states that "Funding sources should be considered in the risk-of-bias assessment conducted for systematic reviews that are part of an IRIS assessments." (Ref: [National Research Council. Review of EPA's Integrated Risk Information System \(IRIS\) Process](#). Page. 79. Washington, DC: National Academies Press; 2014).
3. The authors should consider an expanded and more robust definition of NAMs, as suggested by the NAS in a [recent report](#).
4. The authors should also consider expanding the participant affiliation criteria to include NGOs (section 2.2.2.1).

Suggestions for addressing the comments: Authors should consider expanding their introduction to give a more detailed description of the development of INVITES-IN, potentially including clarification as to whether financial conflicts of interest would be considered within the currently described risk domains. The inclusion of experts in developing or validating in silico approaches such as QSARs, Read-across etc. should also be considered when recruiting participants, including those who may be employed by NGOs.

3. Is the methodology and analysis pipeline sound (and, if appropriate, statistically powered)?

☐ Strongly disagree	☒ Somewhat disagree	☐ Neither agree nor disagree	☐ Somewhat agree	☐ Strongly agree
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Comments: Overall, the methods and analysis are sound and statistical power has been addressed. Nonetheless, the reviewers had some questions related to the selection of participants and noted some issues particularly in relation to assessing, analysing and reporting the accuracy of the tool.

Reviewer 1

1. The aim was to test of the performance of INVITES-IN. By performance, they mean the extent to which results of using INVITES-IN are the same for different users (consistency), the amount of time and cognitive effort it takes to apply INVITES-IN (assessor workload), the precision and potential for systematic error in results of applying INVITES-IN (accuracy), and how easy it is to use INVITES-IN (user experience). My comment is about the accuracy where details are missing in such protocols. A. I feel the authors should use checklist to consider important steps in methodology dealing with accuracy (e.g. [Stard](#)), especially with B. blinding/blindness that should be detailed, C. statistical analyses that should include AUO/DOR/ sensitivity, specificity, predictive values, and likelihood ratios by combining what is true and positive/negative and confidence interval. This is important because the authors should clearly mention on (D) what extent the tool will be consider accurate enough (e.g. 80% for Sensitivity/ Specificity). I strongly not recommended statistical test based (p significant), since association with gold standard is different to be predict low bias if positive/negative.

Reviewer 3

1. The authors should explicitly define what a project group is.
2. For validation purposes, the authors should require participants in the user testing to disclose any financial conflicts of interest for any of the studies being evaluated (Section 2.2.2.1).
3. Under data analysis and reporting, the authors should also consider collecting information on participants' race/ethnicity.

Suggestions for addressing the comments: Authors provide an explicit definition of project group, and consider inclusion of other important information about recruited test users including ethnicity and potential conflicts of interest. The applicability of [the STARD 2015 guidelines](#) (DOI: 10.1136/bmjopen-2016-012799) to testing the accuracy of a risk-of-bias tool for in vitro methods should be considered and discussed. More detail on blinding should be given where appropriate. Finally, the statistical analysis of the accuracy of the tool should consider the applicability of statistics commonly reported for this type of study such as area under the receiver operating characteristics curve, diagnostic odds ratio, specificity, sensitivity etc. as well as state aims in terms of specificity and sensitivity of the tool, offering a rationale for their balance between false positives and false positives.

4. Are the methods detailed and clear enough to permit replication of the experimental procedures and analysis pipeline?

☐ Strongly disagree	☒ Somewhat disagree	☐ Neither agree nor disagree	☐ Somewhat agree	☐ Strongly agree
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Comments: The clarity of the description of specific parts of the methods could be improved to ensure replicability, notably: the selection of users to test the tool, the methods for the systematised review and power simulations.

Reviewer 1

1. Clarity of the systematized review (not systematic here in my opinion) and how will be used here might be clarified.

Reviewer 2

1. What were the criteria for “scientists having good experience with *in vitro* methods”? Only experimenters? Some more details about the “experience space” would be welcome (besides the eligibility criteria in Table 2).
2. Power simulations: “*The rating were therefore correct for 83.8% of the studies. This corresponded to an ICC of approximately 0.65*”. Please provide the mathematical equation. The whole description on page 16 is unclear as no formal mathematical description of the simulation and assessment is provided, e.g. the term “95% confidence interval” has not been specified (I guess the percentiles of the simulation outcome distribution are meant), and “95% margin of error for estimating ICC” is rather unclear in the context of simulations.
3. I found it rather hard to follow the description for the power simulations (especially, which & how the random generator is used), and would suggest to provide the corresponding calculation code(s) as suppl. material. (assuming R was used?).

Suggestions for addressing the comments: When not previously addressed (i.e. expertise of potential users), the required changes are editorial in nature and should only require the addition of further details on methods.

5. Have the authors specified sufficient outcome-neutral tests for ensuring the results obtained can test the stated hypotheses?

☐ Strongly disagree	☐ Somewhat disagree	☒ Neither agree nor disagree	☐ Somewhat agree	☐ Strongly agree
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Comments: This aspect was not commented on by the reviewers and appears to have been adequately addressed in this study protocol.

Suggestions for addressing the comments: None

Other comments

Reviewer 1

1. I should mention that I am not familiar with such In vitro techniques, but I have some experience of epidemiology and clinical toxicology. However, sometimes it is hard to follow as I mentioned before.
2. Details of ICC and numbers of judges and attempt is over developed and should be mentioned in supplemental, with keeping the main attended methods and results
3. Potential limitations of such approaches should be discussed
4. An ethical approval was obtained by Norwegian Institute of Public Health: when? What number?

The structure for the peer-review evaluation is derived from the reviewer and author guidelines for Registered Reports (Chambers et al. 2023, <https://osf.io/pukzy>).

Editorial Evaluation, Round 1

Basic information checklist

Passed?	Item checked
Y	Preprint and supplemental material is available on Zenodo
N	Full name, affiliations, and ORCIDs (if available) of each author are present
N	Structured abstract has been provided
Y	Author contributions in the form of a CRediT statement have been given
Y	Complete funding details are given, or “no funding” is declared
N	An informative disclosure of interests for all authors has been provided, or disclosure refused
Y	3rd-party data and program code are fully cited, if relevant (TOP 1, level 3)
N	Supplemental materials have plain English file names

Editor's Comments

Please provide authors ORCID numbers if available.

The abstract is rather brief, please provide a structured abstract.

Not all authors have provided an informative disclosure of interest form, could all authors please disclose their interests (or explicitly refuse to do so).

There are some supplemental materials in the manuscript file as well as a separate excel file. Please create separate files with plain English file names for all supplemental materials.

Triage Checks

Quick Links: [Objectives](#) | [Experiment](#) | [Analysis](#) | [Transparency](#) | [Ethics](#)

Domain	1. Objectives, context, and mode of research				
Level of Concern	☐ Critical	☐ Serious	☐ Moderate	☐ Minor	☒ None
Specific Issues	☐ Clarity about the knowledge goals or hypotheses of the research (e.g. PECO) ☐ Justification for conducting the research (context, rationale) ☐ Differentiation between research in confirmatory or exploratory mode ☐ Other (please explain in comments)				
Comments	The objectives of the research are clear, as are the context in which this protocol has been developed.				

Quick Links: [Objectives](#) | [Experiment](#) | [Analysis](#) | [Transparency](#) | [Ethics](#)

Domain	2. Soundness and rigour of planned experimental approach				
Level of Concern	☐ Critical	☐ Serious	☐ Moderate	☐ Minor	☒ None
Specific Issues	☐ Theory as to how the study delivers the knowledge goal ☐ Planned identification and authentication of study components, if applicable ☐ Planned approach to power and replicates ☐ Plan for bias controls (randomisation, masking, calibration, training, etc.) ☐ Other (please explain in comments)				
Comments	No issues with the soundness and rigour of the planned experimental approach identified at triage stage. Detailed description of statistical methods including power analyses.				

Quick Links: [Objectives](#) | [Experiment](#) | [Analysis](#) | [Transparency](#) | [Ethics](#)

Domain	3. Plan for the analysis of data and reporting of results				
Level of Concern	☐ Critical	☐ Serious	☐ Moderate	☒ Minor	☐ None
Specific Issues	☐ Planned data normalisation and cleansing techniques ☒ Planned analysis pipeline (qualitative, narrative, and/or statistical methods) ☐ Planned summarisation and visualisation of results ☐ Selectivity in use of data or emphasis on study results ☐ Other (please explain in comments)				
Comments	A detailed plan for data analysis and reporting is given. It does not appear to include analyses of the influence of the characteristics of selected				

	participants on study outcomes. This is related to minor issue highlighted in transparency domain.
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Quick Links: [Objectives](#) | [Experiment](#) | [Analysis](#) | [Transparency](#) | [Ethics](#)

Domain	4. Transparency and reproducibility				
Level of Concern	☐ Critical	☐ Serious	☐ Moderate	☒ Minor	☐ None
Specific Issues	☐ Citation of data, program code, and methods (TOP 1 - L3) ☐ Plans for provision of code, data, and materials (TOP 2,3,4 - L2) ☐ Appropriateness and/or implementation of reporting checklist (TOP 5 - L3) ☒ Other (please explain in comments)				
Comments	Methods are described in detail. The only issue with transparency identified at this stage is related to the identification of study participants to be invited and whether this may introduce bias.				

Quick Links: [Objectives](#) | [Experiment](#) | [Analysis](#) | [Transparency](#) | [Ethics](#)

Domain	5. Ethics and interests				
Level of Concern	☐ Critical	☐ Serious	☐ Moderate	☒ Minor	☐ None
Specific Issues	☒ Informativeness of declarations of interests ☐ Ethical approval for research with human or animal material or participants ☐ Other (please explain in comments)				
Comments	The informativeness of the declarations of interests for all co-authors available as supplementary information is very variable. This should be addressed. Ethical approval has been obtained, no reference number has been provided.				

Summary section

Summary of compliance issues

What is the impact of the compliance issues you have identified on the publishability of the Working Paper?	Do you agree that it is feasible to overcome the compliance issues?
☐ Critical ☐ Serious ☐ Moderate ☒ Minor ☐ None	☐ Strongly disagree ☐ Somewhat disagree ☐ Neither agree nor disagree ☐ Somewhat agree ☒ Strongly agree
Guidance note: Seriousness of the issue is independent of effort required to resolve it. For example, not meeting our requirements for citation standards (TOP 1, level 3) may not require much effort to resolve, but would be a critical issue. Critical and serious issues must be resolved prior to peer-review.	Guidance note: Feasibility is contextual, but in general issues that require additional experimental work or analysis of 3 or more months duration, and/or would result in the study being a different unit of contribution, is less feasible. This is a subjective judgement, provided to inform discussion of how to proceed with the Working Paper.

Summary of manuscript, your overall impression, and suggested course of action

This manuscript describes a protocol for user-testing of a risk-of-bias tool for in vitro (cell cultures) studies. The methods include development of a gold standard and user testing comprising of practical exercises and a user survey. This is in accordance with existing and accepted methods to develop such tools. The protocol is very clear, its objectives and context clearly stated, and methods are described in detail. There were very few minor issues identified at triage, including with the informativeness of declaration of interests for some co-authors, and issues around the identification of study participants and inclusion of analyses (qualitative or quantitative) of the influence of participant characteristics on the outcomes of interest. If a reference has been allocated to ethical approval, this should be provided. These minor issues can be addressed following peer review and this manuscript can progress to peer-review.